

ERQ — Exposure Recoverability Quality

Formal Definition

$$ERQ_i \in [0, 1]$$

ERQ measures:

the degree to which a position structure preserves future maneuverability under changing market geometry.

It is NOT:

- expected return,
- alpha,
- Sharpe,
- predictive probability.

It is a:

- recoverability topology measure.

Core Interpretation

ERQ	Meaning
0	recoverability-destructive exposure
1	recoverability-preserving exposure

High ERQ means:

- bounded damage,
- preserved optionality,
- corridor robustness,
- future re-entry remains admissible.

Low ERQ means:

- forced traversal risk,
- leverage fragility,
- corridor collapse exposure,
- maneuverability destruction.

Philosophical Definition

ERQ answers the question:

“If the regime changes unexpectedly, does this exposure preserve the system’s ability to continue steering?”

This is the true MRTP interpretation.

Formal Structure

ERQ is composed of six sub-components:

$$ERQ_i = \left(\prod_{k=1}^6 erq_k \right)^{1/6}$$

Geometric mean.

Meaning:

- one catastrophic property can collapse the score.

This is intentional.

Because:

- one irrecoverable feature can destroy maneuverability.

The Six Components

A. Forced Traversal Risk

$$erq_A$$

Question:

does this exposure require future market cooperation to exit safely?

Low ERQ:

- naked shorts,
- margin-sensitive leverage,
- forced unwind dependency.

High ERQ:

- bounded optionality,
- unlevered structures,
- no mandatory future traversal.

B. Corridor Width Preservation

$$erq_B$$

Question:

does this exposure preserve multiple future admissible trajectories?

Low ERQ:

- concentrated path dependency,
- narrow positioning.

High ERQ:

- optional structures,
- diversified maneuverability,
- multiple reachable futures.

C. Liquidity Survivability

erq_C

Question:

can this structure survive liquidity deformation?

Low ERQ:

- illiquid exposure,
- squeeze vulnerability,
- unwind fragility.

High ERQ:

- deep liquidity,
- scalable exit capacity,
- stable execution corridor.

D. Boundedness

erq_D

Question:

is maximum damage structurally capped?

Low ERQ:

- theoretically unlimited loss,
- nonlinear deleveraging risk.

High ERQ:

- bounded downside,
- predefined risk envelope.

Example:

- puts > shorts.

E. Re-entry Flexibility

erq_E

Question:

after failure, can the system re-enter efficiently?

Low ERQ:

- catastrophic impairment,
- capital destruction,
- trapped exposure.

High ERQ:

- quick recoverability,
- preserved optionality,
- retained steering.

F. Correlation-Rupture Robustness

erq_F

Question:

does the exposure survive transition-law deformation?

Low ERQ:

- assumes stable correlations,
- collapses under regime shift.

High ERQ:

- structurally adaptive,
- robust under topology change.

Examples

Exposure	ERQ
Naked short	low
Levered concentrated momentum	low
Unbounded volatility selling	very low
Levered beta chase	low
Long put	high

Exposure	ERQ
Bounded call spread	high
Diversified dispersion	medium-high
Optionality-preserving momentum	medium-high
Cash reserve	maximal

M RTP Connection

ERQ enters directly into:

$$MRTP_i = \alpha R_i + \beta C_i - \gamma F_i - \delta P_i + \eta ERQ_i$$

Meaning:

M RTP no longer rewards:

- convexity uniformly.

It rewards:

recoverable convexity.

Runtime Interpretation

High ERQ exposure:

- may expand during widening corridors.

Low ERQ exposure:

- contracts early under fragmentation.

Thus:

- the system governs not merely direction, but:
- topology of exposure itself.

Real-Time Meaning

ERQ is NOT:

- hindsight labeling.

It is:

runtime maneuverability estimation.

The system asks continuously:

"If current market geometry deforms suddenly, does this exposure preserve future admissibility?"

That is the essence of ERQ.

Deepest Interpretation

Traditional finance asks:

"How much can this position make?"

ERQ asks:

"Will this position preserve the ability to continue navigating if the world changes?"

That is the PolygonEye / MRTP distinction.